

ADA STANDARDS OF CARE - 2025

Meeta Bhalla, MD
Tulane Endocrinology





Disclosures

- None

Objectives

- Understand core areas of diabetes management as per ADA standards of care
- Define standards of care as related to core areas of diabetes management
- Describe latest updates to ADA standards of care

THE JOURNAL OF CLINICAL AND APPLIED RESEARCH AND EDUCATION

JANUARY 2025

Diabetes Care.

JANUARY 2025 | VOLUME 48 | SUPPLEMENT 1
DIABETESJOURNALS.ORG/CARE

Diabetes Care.

VOLUME 48 | SUPPLEMENT 1 | PAGES XX-XX



Standards of Care in Diabetes 2025

 American
Diabetes
Association.
ISSN 0149-5992

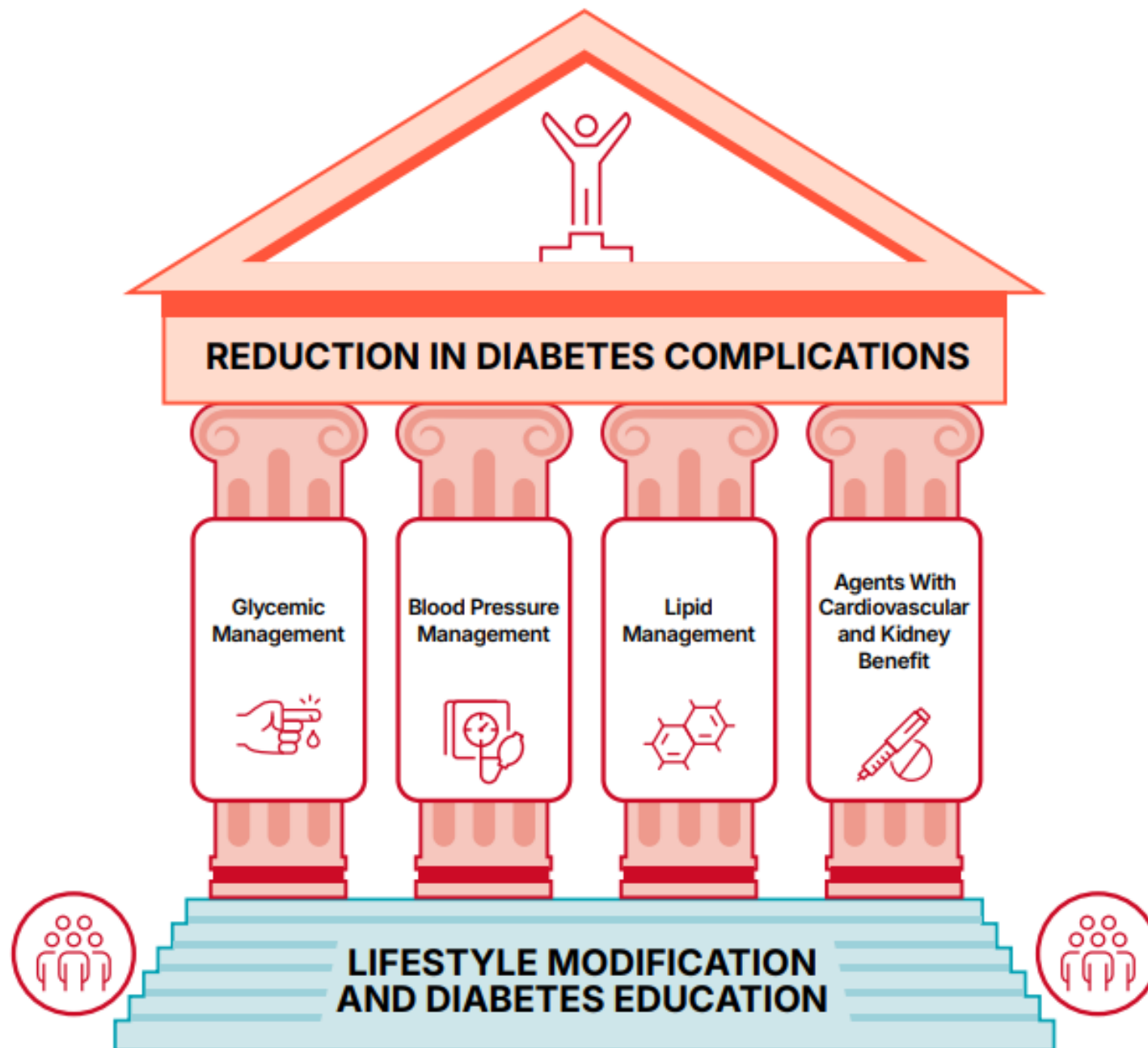
Evidence-Grading System

Table 1—ADA evidence-grading system for “Standards of Care in Diabetes”

Level of evidence	Description
A	Clear evidence from well-conducted, generalizable randomized controlled trials that are adequately powered, including: • Evidence from a well-conducted multicenter trial • Evidence from a meta-analysis that incorporated quality ratings in the analysis Supportive evidence from well-conducted randomized controlled trials that are adequately powered, including: • Evidence from a well-conducted trial at one or more institutions • Evidence from a meta-analysis that incorporated quality ratings in the analysis
B	Supportive evidence from well-conducted cohort studies, including: • Evidence from a well-conducted prospective cohort study or registry • Evidence from a well-conducted meta-analysis of cohort studies Supportive evidence from a well-conducted case-control study
C	Supportive evidence from poorly controlled or uncontrolled studies, including: • Evidence from randomized clinical trials with one or more major or three or more minor methodological flaws that could invalidate the results • Evidence from observational studies with high potential for bias (such as case series with comparison with historical controls) • Evidence from case series or case reports Conflicting evidence with the weight of evidence supporting the recommendation
E	Expert consensus or clinical experience

Introduction:

Standards of Care in Diabetes - 2025. Diabetes Care 2025;48(Suppl. 1):S1-S5



Diagnostic Criteria

	Pre-Diabetes	Diabetes
Hemoglobin A1C	5.7–6.4%	≥6.5%
Fasting plasma glucose	100-125 mg/dL	≥126 mg/dL
2h-plasma glucose during OGTT	140-199 mg/dL	≥200 mg/dL
Classic symptoms+Random plasma glucose	-	≥200 mg/dL

Glycemic Goals

Table 6.3—Summary of glycemic recommendations for many nonpregnant adults with diabetes

A1C	<7.0% (<53 mmol/mol)*†
Preprandial capillary plasma glucose	80–130 mg/dL* (4.4–7.2 mmol/L)
Peak postprandial capillary plasma glucose‡	<180 mg/dL* (<10.0 mmol/L)

Diagnosis and Prevention updates

- Classify people with hyperglycemia into **appropriate diagnostic categories** to aid in personalized management. **E**
- Autoantibody-based **screening for presymptomatic T1D** should be offered to those with family history of T1D or otherwise known elevated genetic risk. **B**

Diagnosis and Prevention updates

- Sleep is a **central component** in the management of prediabetes and T2D
 - At par with other lifestyle behaviors – nutrition and PA
 - U-shaped association between sleep duration (<6 h or >9 h) and T2D incidence with up to 50% increase in risk of T2D, including progression from prediabetes

Healthy lifestyle behaviors, DSMES, SDOH

Goal: Cardiorenal Risk Reduction in High-Risk Patients with Type 2 Diabetes

Goal: Achievement and management of glycemic & weight management goals

+ASCVD

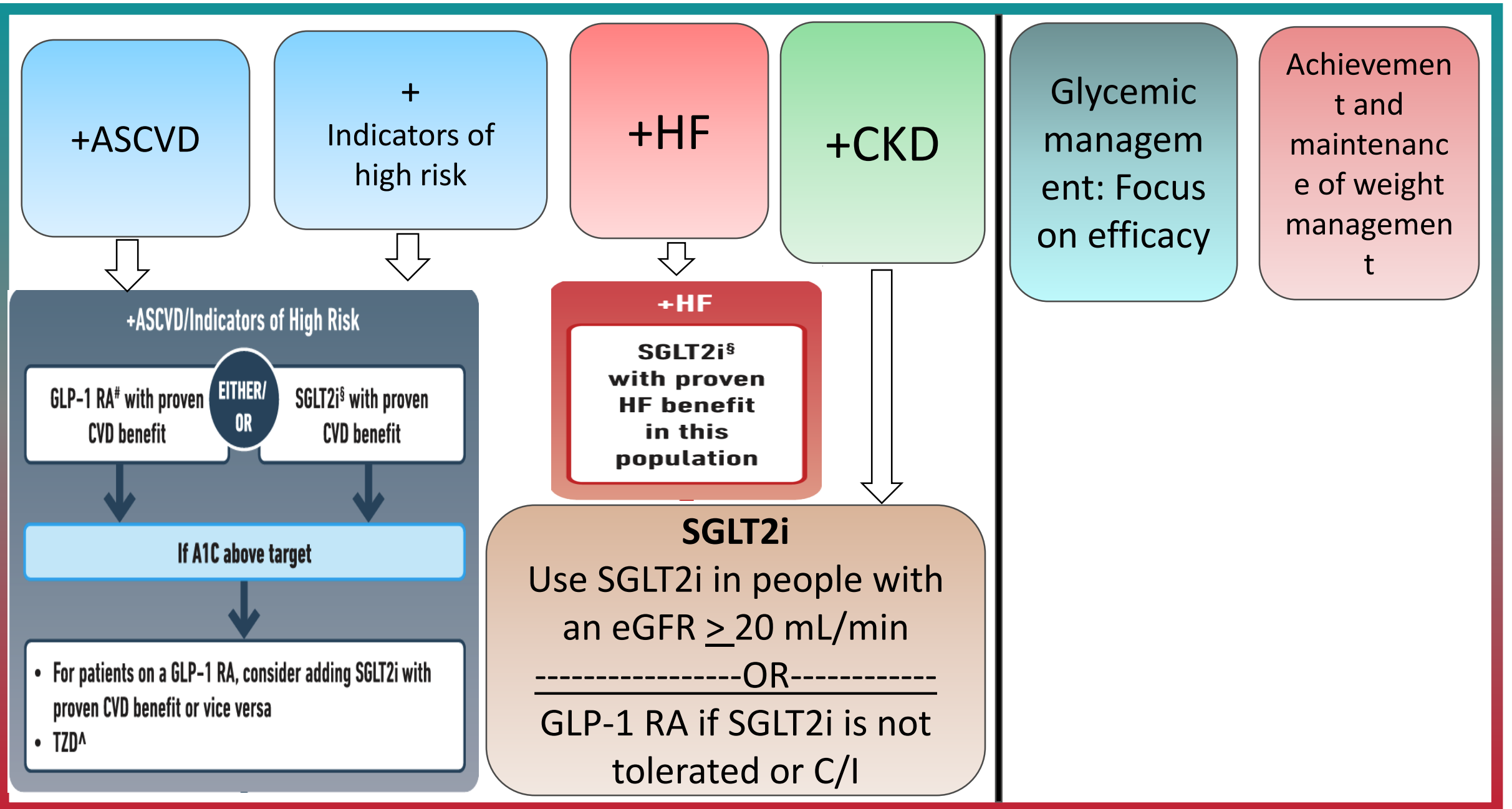
+
Indicators
of high risk

+HF

+CKD

Glycemic
management:
Focus on
efficacy

Achievement
and
maintenance of
weight
management



+ASCVD

+
Indicators
of high
risk

+HF

+CKD

Mitigating risk of MASLD or
MASH: Agents with potential
benefits – GLP-1 RA, dual
GIP/GLP-1 RA, GLP-1 RA
+Pioglitazone
Use insulin in
decompensated cirrhosis

Glycemic
management: Focus
on efficacy

Very High
Dulaglutide, Semaglutide,
Tirzepatide,
Insulin, Combination
injectable (GLP-1
RA/Insulin)
High
GLP-1 (Not listed above),
MTF, SGLT-2i, SU, TZD
Intermediate
DPP-4i

Achievement and
maintenance of weight
management

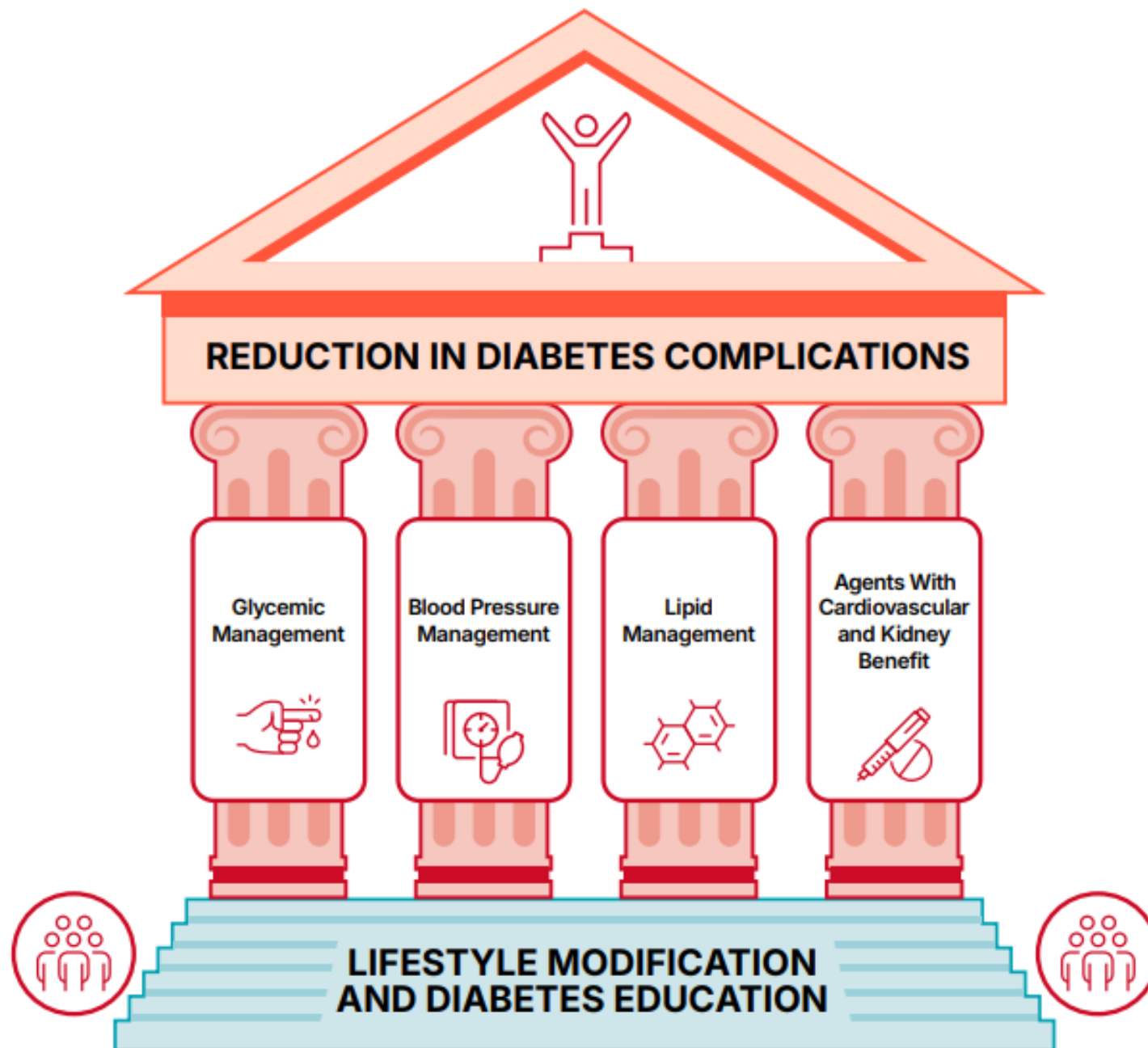
Very High
Semglutide, Tirzepatide
High
Dulaglutide, Liraglutide
Intermediate
SGLT2i, GLP-1 RA (not
listed above)
Neutral
DPP-4i, Metformin

Nutrition and Physical Activity updates

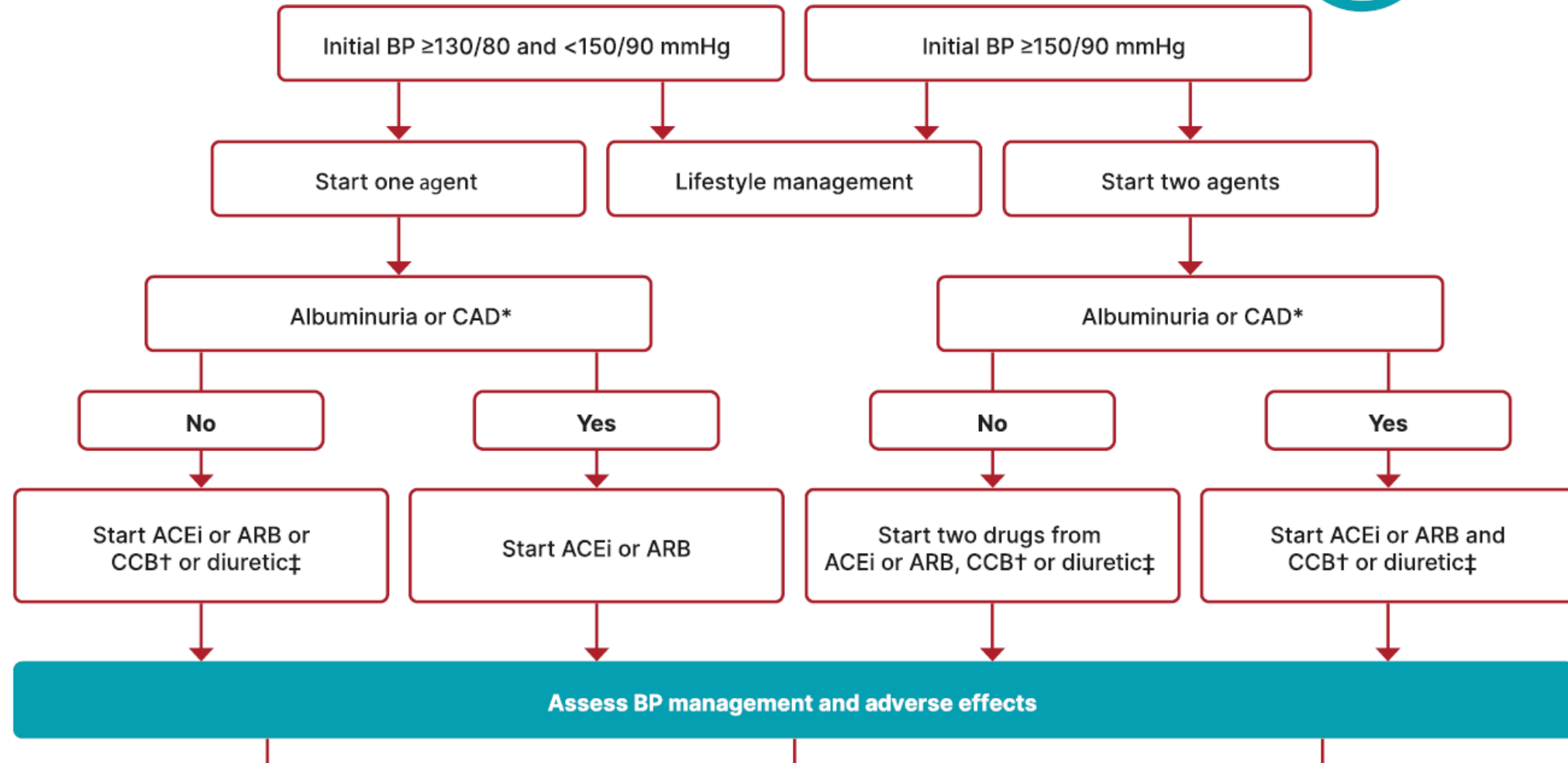
- Individuals treated with SGLT2i should:
 - Avoid **ketogenic eating** pattern
 - Educated on **signs of ketoacidosis** and methods of risk mitigation
 - Appropriate **tools for ketone measurement**
 - Avoid **fasting** and maintain appropriate insulin therapy **E**

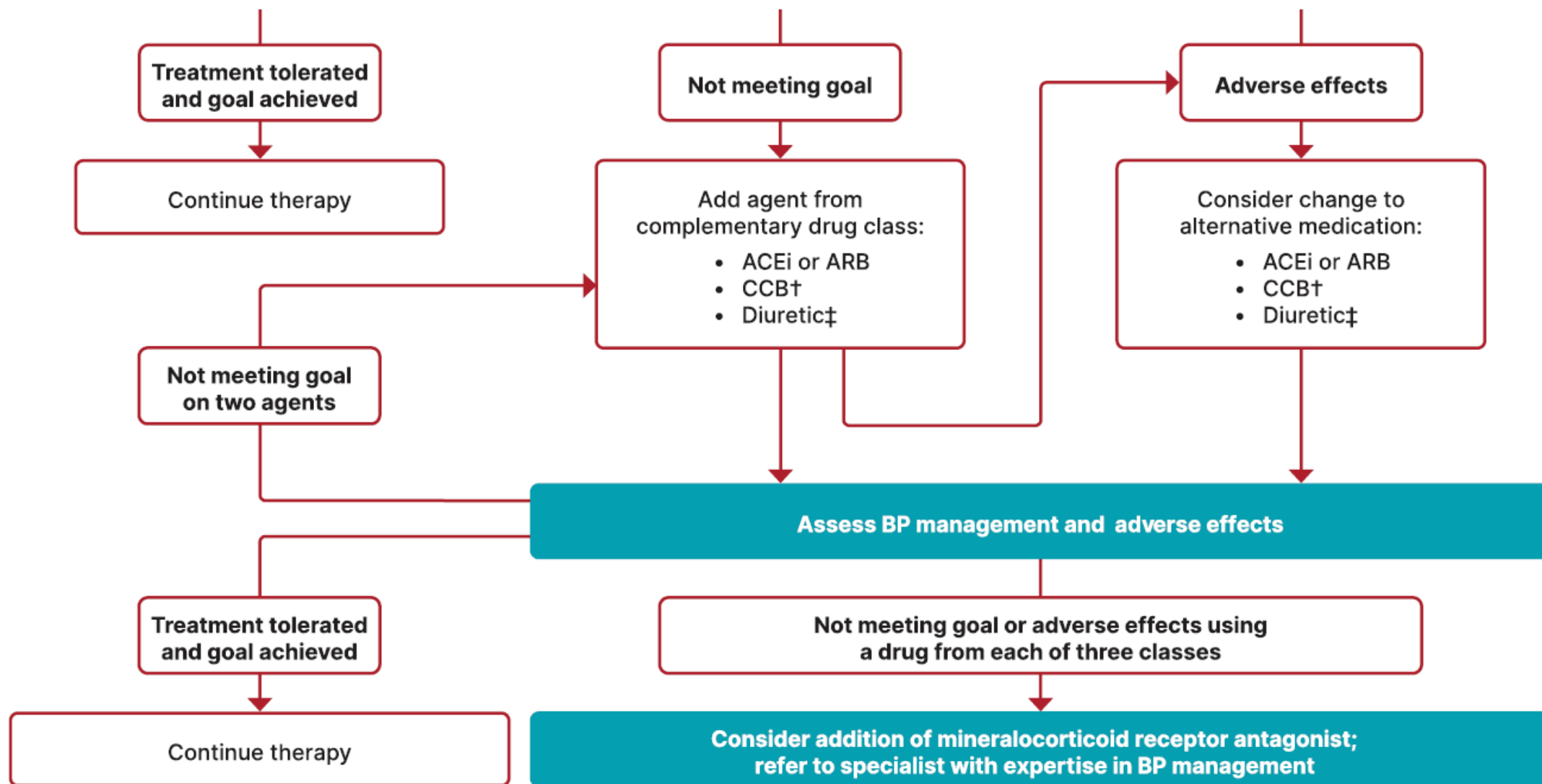
Nutrition and Physical Activity updates

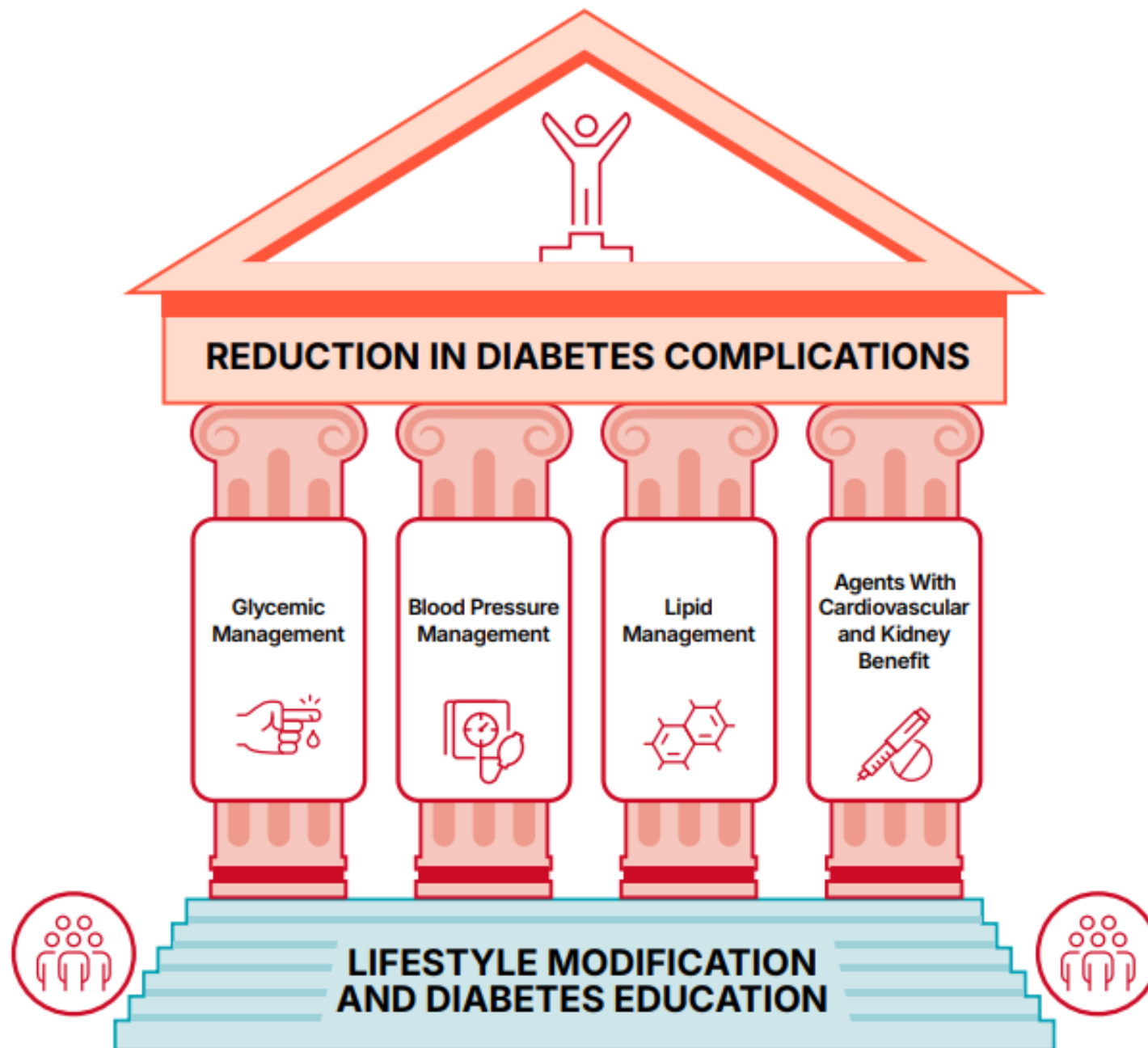
- Screen for malnutrition in those at high risk of malnutrition - weight management pharmacotherapy B or metabolic surgery A
- Counsel adults and youth treated with weight management pharmacotherapy or metabolic surgery regarding muscle strengthening exercises, for maintaining lean body mass. E



Recommendations for the Treatment of Confirmed Hypertension in Nonpregnant People With Diabetes







Lipid Management

- Age 40-75 y + DM + **high CV risk** → high intensity statin to
 - 1. ↓ LDL by >50% and
 - 2. target LDL **<70 mg/dL**
 - If not at goal, add ezetimibe or PCSK9 inhibitor to maximum tolerated statin
- Age 40-75 y + DM + **CAD** → high intensity statin to
 - 1. ↓ LDL by >50% and
 - 2. target LDL **<55 mg/dL**
 - If not at goal, add ezetimibe or PCSK9 inhibitor to maximum tolerated statin

Lipid Management

- Intolerance to statin therapy: Bempedoic acid
- May not be statin intolerant: switching to another agent, starting at a lower dose or alternate day therapy regimen.

Lipid Management for Primary Prevention of Atherosclerotic Cardiovascular Disease Events in People With Diabetes in Addition to Healthy Behavior Modification

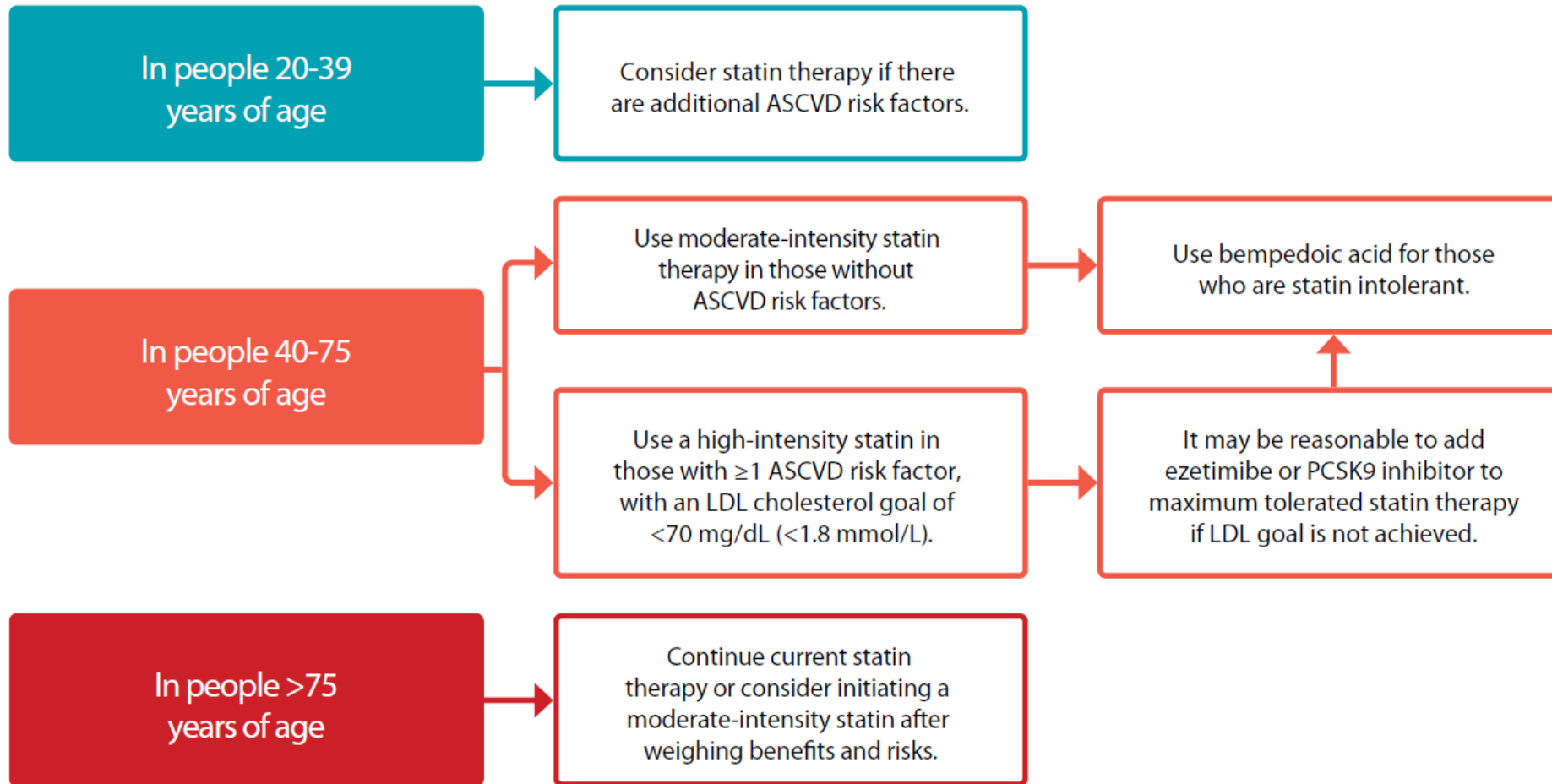
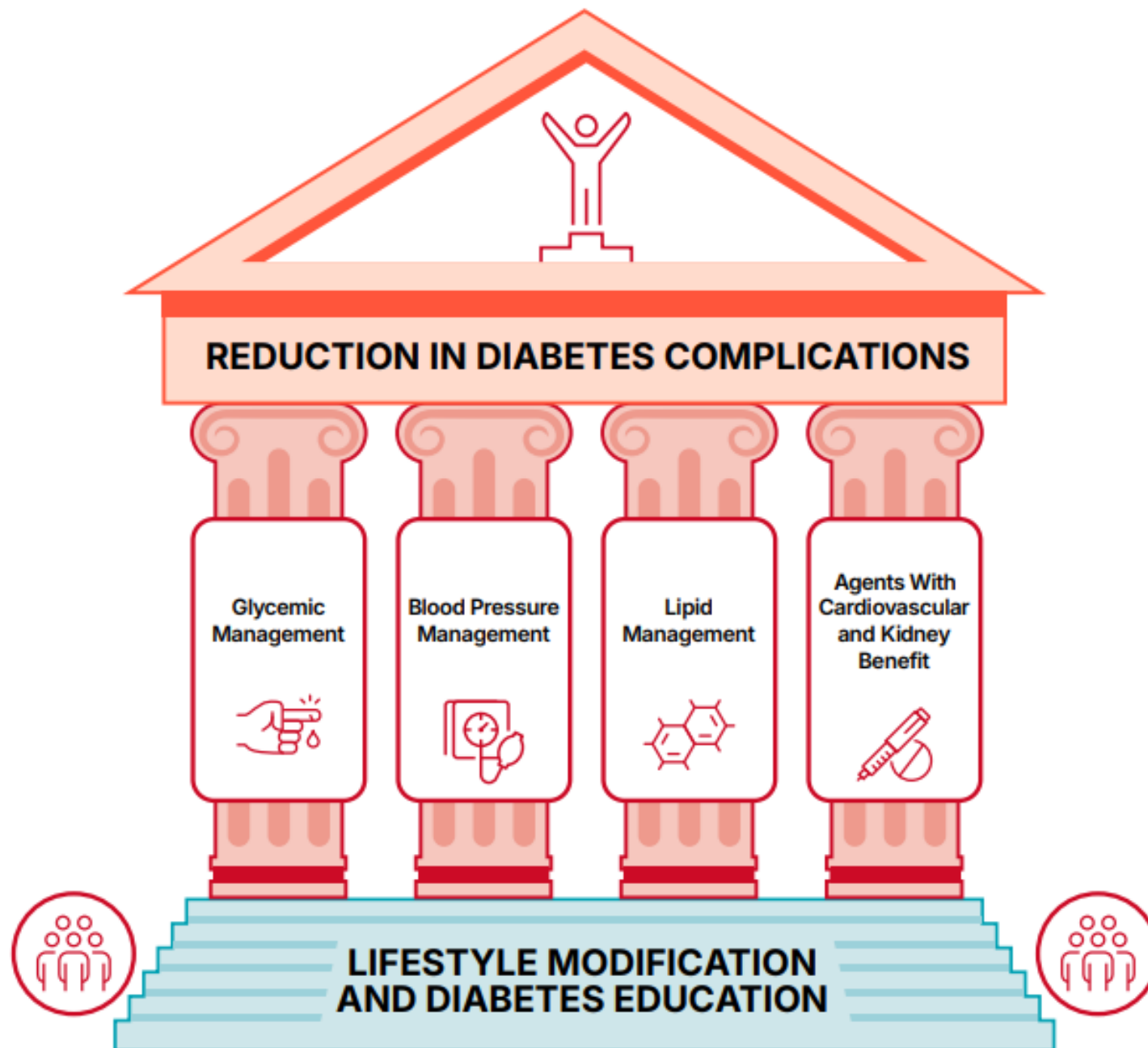


Figure 10.3—Recommendations for primary prevention of atherosclerotic cardiovascular disease (ASCVD) in people with diabetes using cholesterol-lowering therapy. Adapted from “Standards of Care in Diabetes—2024 Abridged for Primary Care Professionals” (325).



Cardiovascular Disease Screening

- Consider screening adults with diabetes with natriuretic peptide (B-type natriuretic peptide [BNP]) or N-terminal pro-BNP [NTproBNP] to facilitate prevention of stage C heart failure.
- In asymptomatic individuals with diabetes and abnormal natriuretic peptide levels, echocardiography is recommended to identify stage B heart failure.

Cardiovascular Disease Screening

- Age ≥ 50 y + Diabetes+ microvascular disease in any location/foot complications/end-organ damage: Screen for PAD with ankle-brachial index.
- Diabetes duration ≥ 10 years: Consider screening for PAD.

Cardiovascular disease - Treatment

- Diabetes + multiple ASCVD risk factors or established ASCVD:
 - SGLT2 inhibitor or GLP-1 RA with demonstrated CV benefit to ↓ risk of major adverse CV events

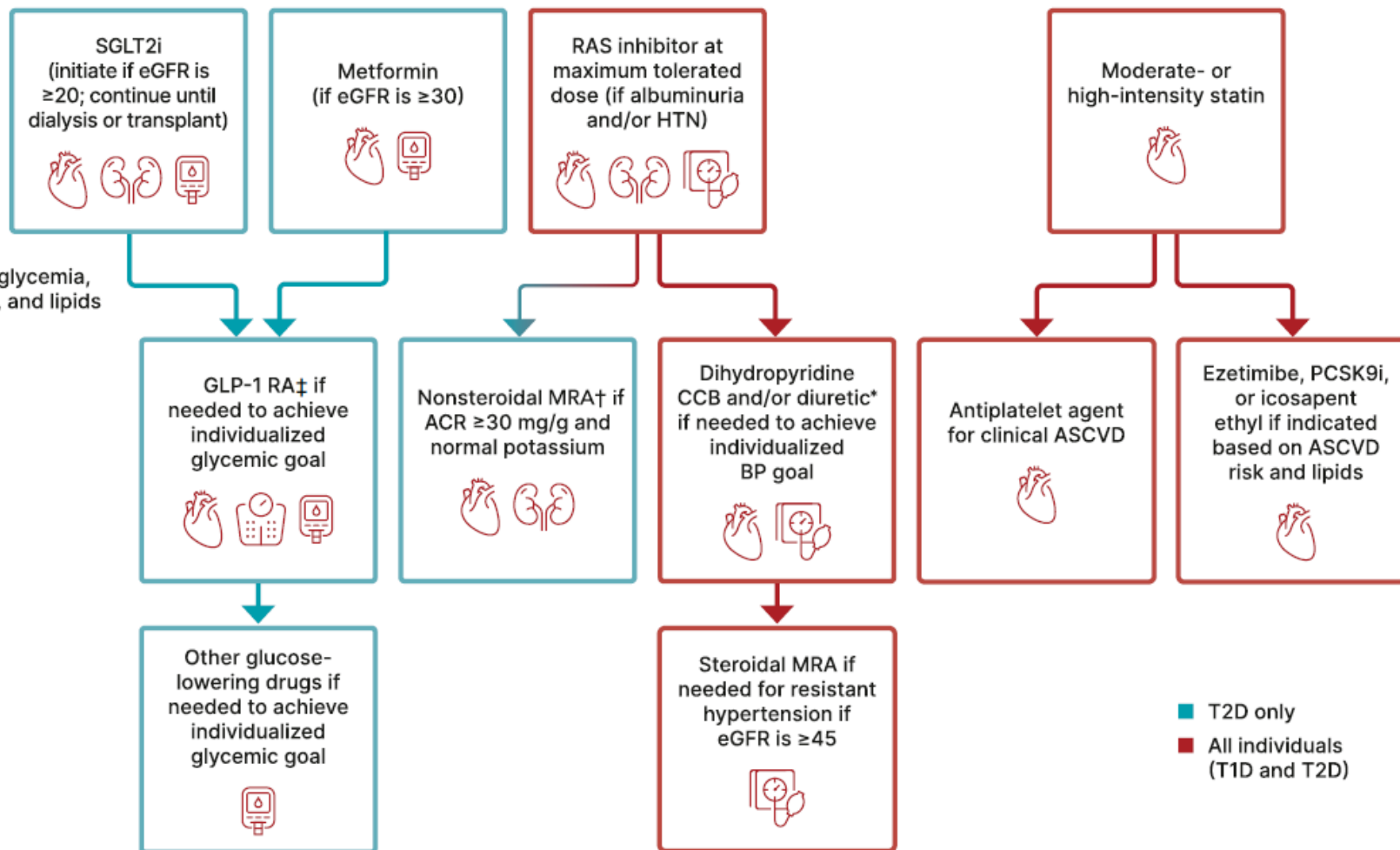
Cardiovascular disease - Treatment

- Diabetes + heart failure with either preserved or reduced ejection fraction: **SGLT2 inhibitor** with proven benefit to **improve symptoms, physical limitations, heart failure hospitalization and quality of life.**

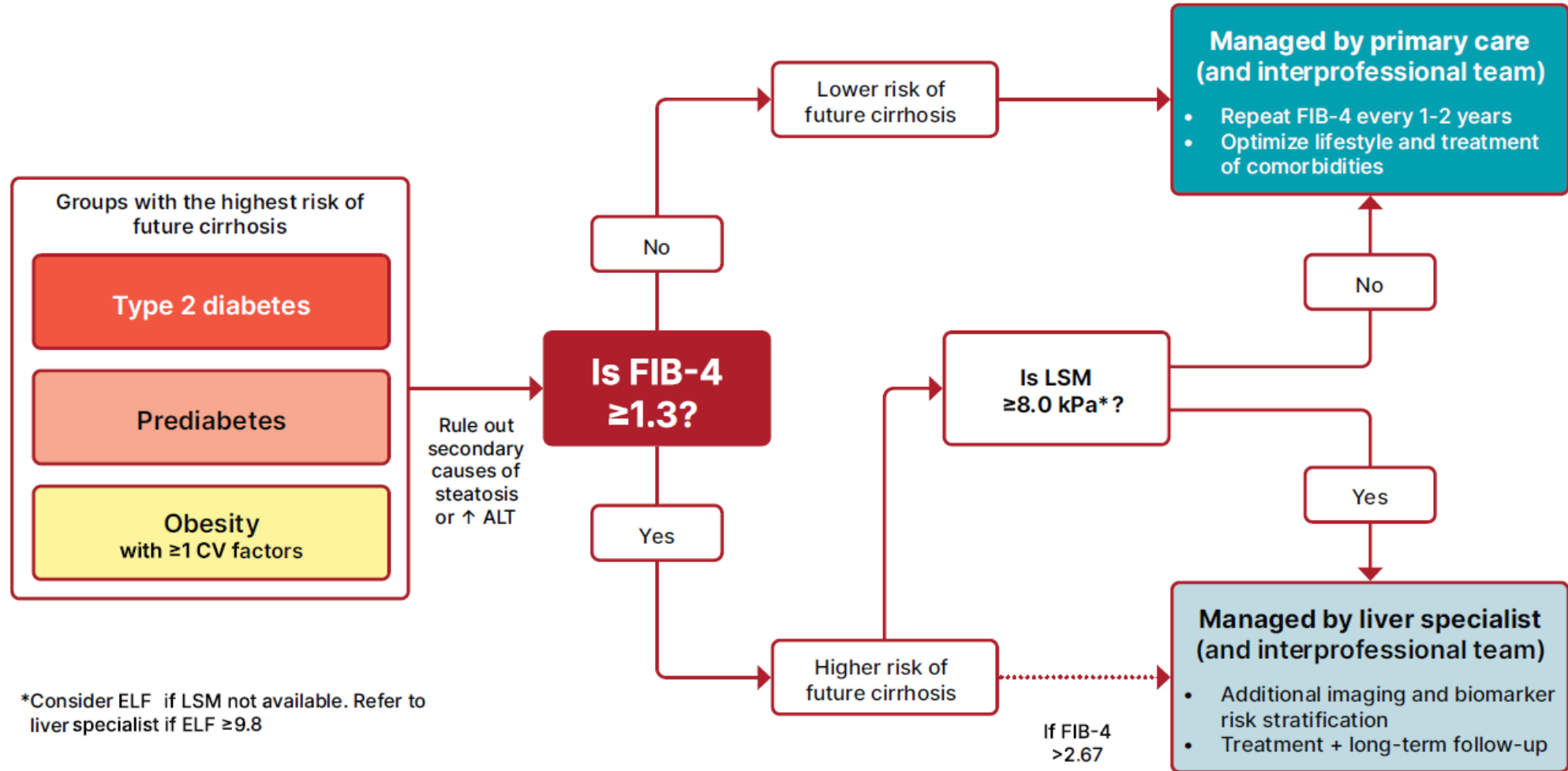
LIFESTYLE



FIRST-LINE DRUG THERAPY

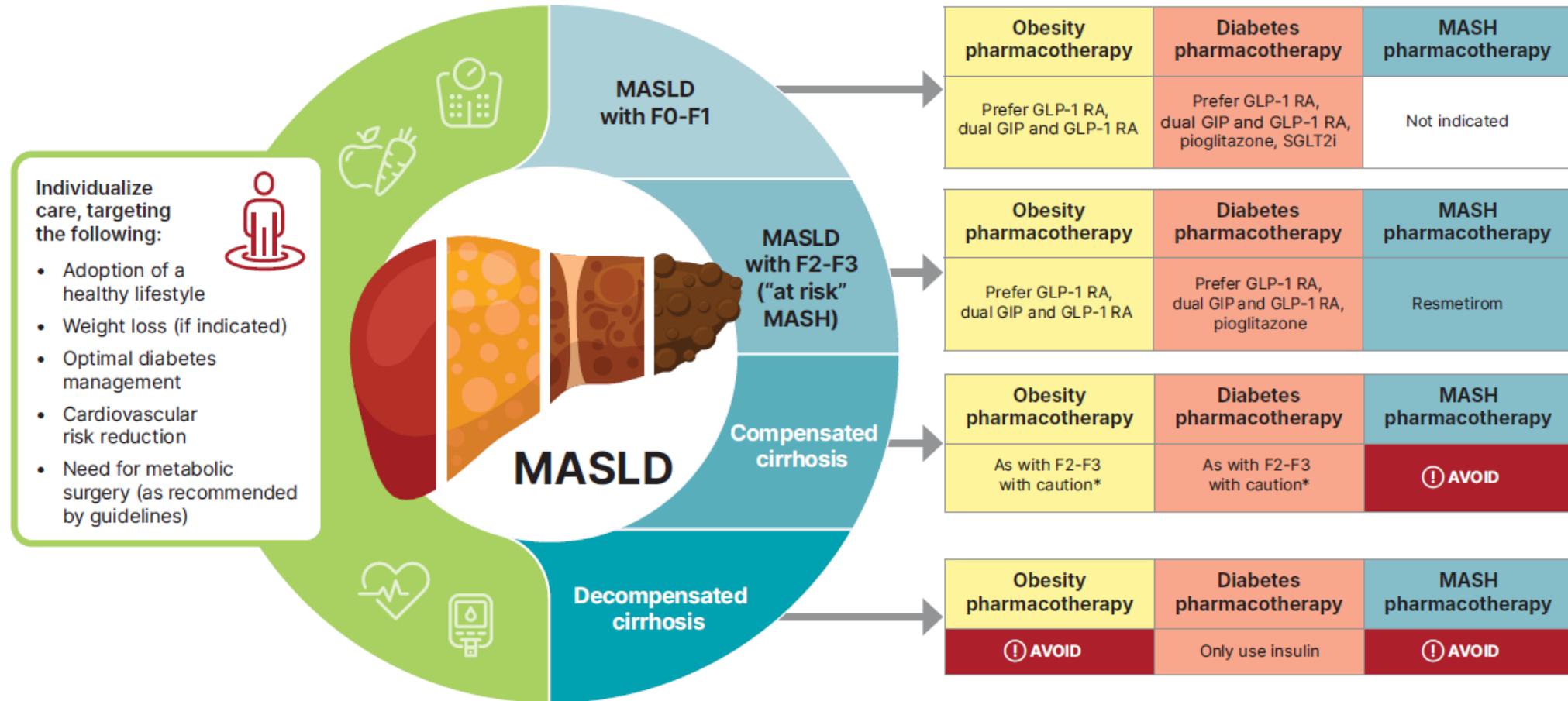


Diagnostic Algorithm for the Prevention of Cirrhosis in People With Metabolic Dysfunction-Associated Steatotic Liver Disease (MASLD)



*Consider ELF if LSM not available. Refer to liver specialist if ELF ≥ 9.8

Metabolic Dysfunction–Associated Steatotic Liver Disease (MASLD) Treatment Algorithm



*Individualized care and close monitoring needed in compensated cirrhosis given limited safety data available.

Dental care

- People with diabetes should be referred for a dental exam **at least once per year. E**
- Coordinate efforts between medical and dental teams to adjust glucose-lowering medication prior to and in the post–dental procedure period as needed. **B**

Sexual Dysfunction

- In women with diabetes or prediabetes, **inquire about sexual health** by screening for libido, arousal, and orgasm difficulties, particularly in those who experience depression and/or anxiety and those with recurrent UTIs. **B**

Sexual Dysfunction

- In postmenopausal women with diabetes or prediabetes, screen for symptoms and/or signs of genitourinary syndrome of menopause, including vaginal dryness and dyspareunia. B

Religious Fasting

- Use International Diabetes Federation and Diabetes and Ramadan International Alliance comprehensive **prefasting risk assessment** to generate a **risk score for safety of religious fasting**. Provide fasting focused education to minimize risks. B

Religious Fasting

- Assess and optimize treatment plan, dose, and timing for people with diabetes well in advance of religious fasting to reduce risk of hypoglycemia, dehydration, hyperglycemia, and/or ketoacidosis.

B

International Diabetes Federation- Diabetes and Ramadan Alliance (IDF-DAR) Fasting Risk Assessment

Stratifies fasting risk during Ramadan in diabetic patients.

MANAGEMENT

- Fasting is a personal decision.
- Provide education on recognizing hypoglycemia and hyperglycemia symptoms.
- Regardless of a patient's risk group, patients should be counseled to break their fast if they experience hypoglycemia episodes.
- Encourage close monitoring of glucose levels during fasting hours. Emphasize that glucose testing during Ramadan does not break the fast.
- Adjustment of the patient's diabetes treatment plan may be required to avoid adverse events during fasting.
- Allow flexibility in glycemic targets during Ramadan.
- Consider a medical nutrition consultation.
- For additional guidance, see the ADA/EASD consensus recommendations for management of diabetes during Ramadan [here](#) or the Diabetes and Ramadan practical guidelines [here](#).

Table 5.5—Changes in medications during fasting

Medication name	Risk of hypoglycemia	Timing	Total daily dose
Metformin, SGLT2 inhibitor, DPP-4 inhibitor, GLP-1 receptor agonist, acarbose, or pioglitazone	Low	• If once daily, then take at main mealtime. • If twice daily, then split dose between the two meals. • If once weekly, no change of time.	• No change
New generation sulfonylurea (glimepiride and gliclazide)	Low to moderate	• If once daily, then take at main mealtime. • If twice daily, then split dose between the two meals.	• Reduce dose if glucose levels are within individualized goal range and if no hypoglycemia or hyperglycemia is present at baseline.
Older generation of sulfonylurea (glyburide)	Moderate to high	• Take at time of main meal	• Replace with newer-generation sulfonylurea or reduce dose by 50%.
Basal insulin	Moderate to high	• For longer-acting basal analogs (glargine 300 or degludec), no need to change timing. • For other basal insulins, take at beginning of breaking fast meal.	• Choose the insulin with lower risk of hypoglycemia among the class. • Reduce dose by 25–35% if not well managed.
Prandial insulin	High	• At mealtime	• Reduce dose of insulin for the meal followed by fasting (35–50%). • For other meals, insulin dose should match carbohydrate intake.
Mixed insulin and insulin coformulations	High	• If once daily, then take at main mealtime. • If twice daily, then split dose between the two meals	• Reduce dose of insulin for the meal followed by fasting (35–50%). • For other meals, no change of dose.

DPP-4, dipeptidyl peptidase 4; GLP-1, glucagon-like peptide 1; SGLT2, sodium–glucose cotransporter 2.

Diabetes Technology

- Consider using rtCGM and isCGM in adults with type 2 diabetes treated with **glucose-lowering medications other than insulin** to achieve and maintain individualized glycemic goals. Choice of device should be made based on the individual's circumstances, preferences, and needs. B

Weight Loss Pharmacotherapy

- Use of compounded products that are not approved by FDA is not recommended due to uncertainty about their content and resulting concerns about safety, quality, and effectiveness. E

Weight Loss Pharmacotherapy

- If glucose-lowering medication is unavailable (e.g., in shortage), it is recommended to **switch to a different FDA-approved medication** (in the same or different class) with similar efficacy, as clinically appropriate. **E**
- Upon resolution of unavailability (e.g., shortage), reassess the appropriateness of resuming original FDA-approved medication. **E**

Diabetes and Pregnancy

- Individuals with diabetes of childbearing potential should be counseled on contraception options **A** and the **impact of some glucose-lowering medications on contraception** efficacy. **C**
 - GLP-1 RA and dual GIP/GLP-1 RA impact levels of oral contraception during time of highest impact on gastric emptying – use second form of contraception until maintenance dose is achieved and used for at least 4 weeks

Diabetes and Pregnancy

- **Preconception planning** should address attainment of glycemic goals, **A** time frame for discontinuing noninsulin glucose lowering medications, **E** and optimal glycemic management in preparation for pregnancy. **A**

Cancer Therapies

- Treatment with immune checkpoint inhibitors for cancer can cause acute autoimmune type 1 diabetes (1% of people) due to rapid onset of insulin deficiency - often presents with DKA and requires insulin therapy

Cancer Therapies

- mTOR kinase inhibitors (e.g., everilomus) can interfere with insulin signaling, leading to **impaired insulin secretion and increased insulin resistance**
- **Metformin is first-line** treatment, with insulin and other noninsulin treatments added in a stepwise fashion dependent on glucose level



Thank you!